Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\left|\sum_{\alpha\beta\gamma\delta}\langle\Phi_\alpha\Phi_\beta\Phi_\gamma\Phi_\delta\rangle_c\;/\;\sum_{\alpha\beta\gamma\delta}\langle\Phi_\alpha\Phi_\beta\Phi_\gamma\Phi_\delta\rangle\right| = \left|\langle\Phi^4(x)\rangle_c\;/\;\langle\Phi^4(x)\rangle\right|\;(n=4)\;,$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\int_0^{2\pi} d\sigma X^i(\sigma)\hat{P}_i(\sigma) = x_0^i\hat{p}_0^i + \frac{g_{ij}}{2\sqrt{\alpha'}}\sum_{n=1}^{\infty}\left\{\left(\chi_n^i\alpha_{-n}^j + \bar{\chi}_n^i\tilde{\alpha}_{-n}^j\right) + \left(\bar{\chi}_n^i\alpha_n^j + \chi_n^i\tilde{\alpha}_n^j\right)\right\}\;.$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\left(\phi_1(x), \begin{pmatrix}\phi_2(x)\\ \phi_3(x)\end{pmatrix}, \begin{pmatrix}\phi_4(x)\\ \phi_5(x)\end{pmatrix}\right) \to \left(c\,e^{\lambda x}\phi_1(x), U\begin{pmatrix}e^{-\lambda' x}\phi_2(x)\\ e^{-\lambda' x}\phi_3(x)\end{pmatrix}, V\begin{pmatrix}e^{\lambda'' x}\phi_4(x)\\ e^{\lambda'' x}\phi_5(x)\end{pmatrix}\right)$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\frac{-i\langle Ta_{\mathbf{k}}^{\dagger t}(-\mathbf{q})a_{\mathbf{k}'}^{\dagger}(\mathbf{q}')\rangle}{\langle T1\rangle} = \sum_n\; exp(\omega_n t)\;\frac{-i\langle Ta_{\mathbf{k}}^{\dagger n}(-\mathbf{q})a_{\mathbf{k}'}^{\dagger}(\mathbf{q}')\rangle}{\langle T1\rangle}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\bar{G}^{(\zeta)}(\beta) = \frac{(\beta+i\pi)\sinh\left(\frac{\pi\beta}{2\pi+\zeta}\right)}{\Gamma\left(1-\frac{\pi-i\beta}{2\pi+\zeta}\right)\Gamma\left(1-\frac{\pi+i\beta}{2\pi+\zeta}\right)}\;,$$